

Chronic Renal Disease Prediction Using Machine Learning Methods

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Abstract—Chronic renal disease (CRD) has become a major problem. A person with chronic renal disease has no proper functionality of his kidneys which may be fatal because of malfunctioning of kidneys. Machine learning techniques are effective in CRD prediction. This work proposes a workflow to predict CRD status based on clinical data, incorporating data preprocessing, a missing value handling method with collaborative filtering and attributes selection. Out of the 9 machine learning methods considered, the decision tree classifier and random forest classifier are shown to result in the highest accuracy and minimal bias to the attributes. The research also considers the practical aspects of data collection and highlights the importance of incorporating domain knowledge when using machine learning for CRD status prediction.

Index Terms—chronic renal disease, machine learning, classification algorithms, decision tree classifier, random forest classifier

I. INTRODUCTION

The kidneys are two reddish-brown bean-shaped organs found in vertebrates. They are located on the left and right in the retroperitoneal space, and in adult humans are about 12 centimeters($4\frac{1}{2}$ inches) in length. [1]. Each kidney has up to a million functioning units called nephrons. These nephrons filter the fluids and sends out the waste fluid in the form of urine. The kidneys perform their life-sustaining job of filtering and returning to the bloodstream about 200 quarts of fluid every 24 hours. About two quarts are removed from the body in the form of urine, and about 198 quarts are recovered. The urine we excrete has been stored in the bladder for any where from 1-8 hours [2]. Chronic kidney disease (CRD) is a major issue worldwide which is a condition characterized by a gradual loss of kidney function over time, 14% of the world population suffer from CRD. Over 2 million people worldwide currently receive treatment with dialysis or a kidney transplant to stay alive, yet this number may only represent 10% of people who need treatment to live. Chronic kidney disease causes more deaths than breast cancer or prostate cancer [3]. The stages of CRD are determined mainly based on calculating estimated glomerular filtration rate (eGFR) which is based on creatinine level [4], gender, race and age. There are five stages of kidney functionality [5]. In stage 1 of CRD there will be mild kidney damage and eGFR will be 90 or higher. In stage 2 of CRD there will be mild loss of kidney function and eGFR will be 60-89. In stage 3a & 3b of CRD there will

be mild to severe loss of kidney function and eGFR will be 30-59. In stage 4 of CRD there will be severe loss of kidney function and eGFR will be 15-19. In stage 5 of CRD there will be kidney failure or close to failure and eGFR will be less than 15 (see Fig. 1).

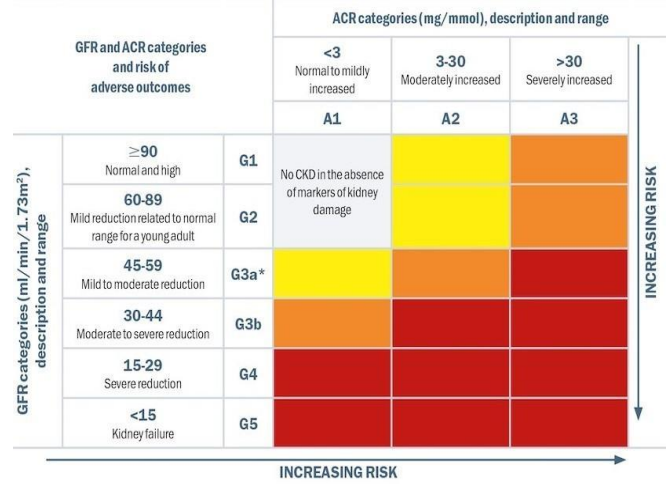


Fig. 1. Heat map of the severity.

To predict positive CRD status and the stages of CRD machine learning can be used. Machine learning uses an effective training data set that is essential to train prediction models that use machine learning algorithms, they extract features most relevant to specified research goals, and to reveal meaningful associations in the data [6]. Application of machine learning methods to predict CRD has been explored based on multiple data sets. Among them, the dataset from UCI repository [7] (referred to as UCI dataset hereafter) is identified as a benchmark dataset [8]–[10]. Similar to most of the related work, this work considers the mentioned benchmark dataset.

When analysing clinical data related to CRD, if there are instances with missing attributes then the missing values handling method should be determined based on the randomness of the way they were missed. Moreover, the UCI data-set [7] has 400 instances which are a comparatively small number of samples with 25 attributes. In this case, the data set may have redundant (highly co-related) features or the data set does not represent all possibilities.

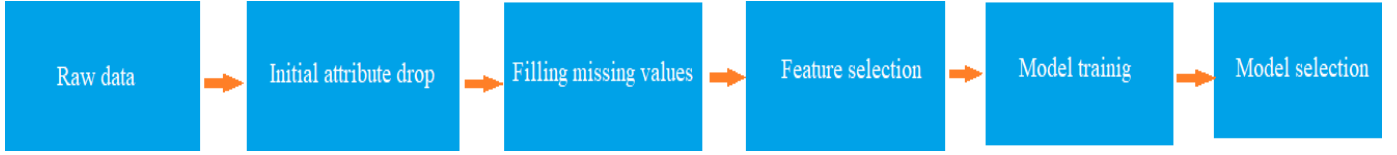


Fig. 2. Proposed workflow.

This work proposes handling the missing values and presents the evaluation of different methods based on this UCI dataset.

Fig 2 describes the proposed workflow of the research that includes collecting data replacing text with numerical data, removing the attributes that contain more than 20% of missing values, using statistical methods the missing values are filled, features have been selected based on both statistical methods and medical importance, 9 models were trained and tuned hyperparameters and finally based on accuracy best model was described.

II. RELATED WORK

Marwa Almasoud, Tomas E Ward have obtained an accuracy of 99.0% in predicting CRD and they said that the three major features that can predict thee CRD are hemoglobin, albumin, specific gravity [8]. MURALE C, KAVINKUMAR P, MANOJ G, SURYA P obtained an accuracy of 100% in prediction of CRD with Logistic regression, KNN classifier and Random forest algorithms and from their proposed work they identified the severity of disease and they stated that person gender and race from which eGFR is calculated is required [9].

Imesh Udara Ekanayake, Damayanthi Herath have obtained 100% of accuracy in prediction of CRD. They have considered hemoglobin, albumin, specific gravity, serum creatinine, hyper tension, diabetes mellitus and blood glucose random as their features that gave them 100% of accuracy. They have used decision tree classifier, random forest classifier, extra tree classifier, ada boost classifier techniques that gave 100% of accuracy out of 11 models [10]. Shanila Yunus Yashfi, Nazmus Sakib, Sadaf Salman Pantho, Md Ashikul Islam, Tanzila Islam, Pritilata, Mohammad Shahbaaz proposed that they have used real time data and data set from UCT and used 2 models in predicting the CRD and got highest accuracy by random forest classifier [11].

Njoud Abdullah Almansour, Hajra Fahim Syed, Nuha Radwan Khayat, Rawan Kanaan Altheeb, Renad Emad Juri, Jamal Alhiyafi, Saleh Alrashed, Sunday O Olatunji proposed prediction of CRD by using Artificial Neural Network (ANN) and Support Vector Machine (SVM). The final models of the two proposed techniques were developed using the best-obtained parameters and features. The empirical results from the experiments indicated that ANN performed better than SVM, with accuracies of 99.75% and 97.75%, respectively [12].

III. METHODOLOGY

The proposed methodology consists of 3 key steps: Data preprocessing, models training and model selection (Fig. 2).

A. Data Preprocessing : Missing Value Handling

In this work, data preprocessing was done in 2 steps. Firstly, the attributes having more than 20% data with missing values were filtered out (see Table I). Accordingly, the set of features, (red blood cells, sodium, potassium, white blood cell count, red blood cell count) is excluded in the analysis. The second step in data preprocessing handled the missing values in the remaining data.

In the pre-processing step, missing values have to be handled based on their distributions to achieve reasonable accuracy. Missing values are handled by replacing mean statistical method which means that the mean of the respective feature column is considered and replaced in the place of missing values. Below snippet shows the data of attributes with the percentage of missing values.

id	0.00
age	2.25
bp	3.00
sg	11.75
al	11.50
su	12.25
rbc	38.00
pc	16.25
pcc	1.00
ba	1.00
bgr	11.00
bu	4.75
sc	4.25
sod	21.75
pot	22.00
hemo	13.00
pcv	17.50
wc	26.25
rc	32.50
htn	0.50
dm	0.50
cad	0.50
appet	0.25
pe	0.25
ane	0.25
classification	0.00

Fig. 3. Missing values of attributes

B. Data Preprocessing : Feature Selection

The absolute values of heat map of correlations of attributes to the class label (Fig. 4) show that specific gravity, albumin, sugar, blood pressure, hemoglobin, puss cell and age have the highest correlations (more than 0.5). Then the secondary attributes packed cell volume, blood glucose random, blood urea and serum creatinine have correlation more than 0.3.

The absolute values of extra tree classifier of feature selection to the class label (Fig. 5) show that specific gravity, hypertension, packed cell volume, diabetes mellitus, hemoglobin, albumin and appetite has importance of more than 0.5.

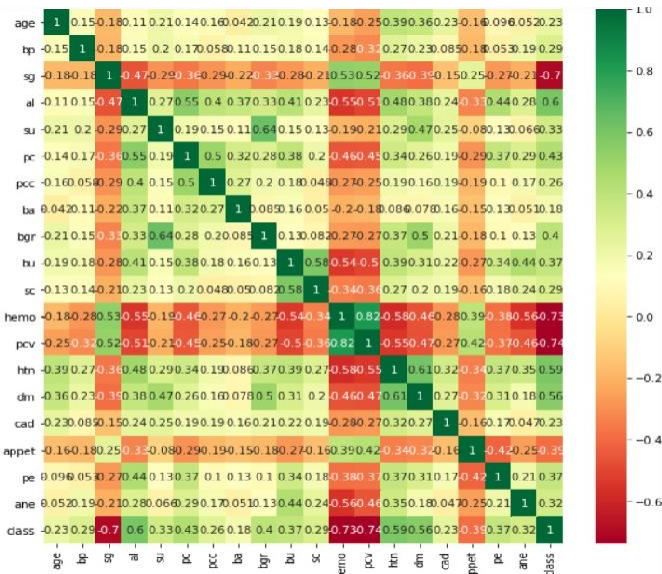


Fig. 4. Heat map of the co-relation of attributes to the class variable

C.

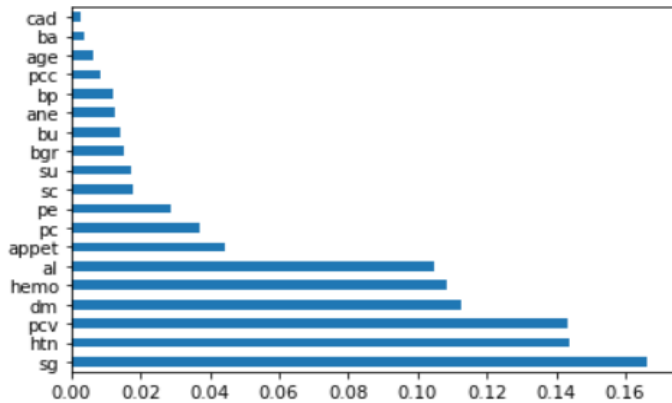


Fig. 5. Extra tree classifier for feature selection to the class variable

Thereafter, considering the distribution of attributes values and the medical perspective of the attributes specific gravity, hemoglobin, hypertension, diabetes mellitus, packed cell volume, albumin, serum creatinine, blood glucose random were selected as the optimal subset of attributes to predict CRD.

C. Model Training

In this work, 9 classification models were considered in training. They are logistic regression, k-Nearest Neighbors (KNN) regression, SVC with a linear kernel, SVC with RBF kernel, Gaussian NB, decision tree classifier, random forest classifier, an ada boost classifier and a Gradient Boosting Classifier. The dataset was divided into 3 parts as 70% training data and 30% test data randomly.

From the mentioned 9 algorithms, 4 algorithms outperformed in training accuracy and testing accuracy. Those are the decision tree classifier, random forest classifier, Gaussian NB, Logistic Regression..

The implementations and evaluation were done using Python Sci-kit, and Keras frameworks.

TABLE I
ACCURACIES OF EACH ALGORITHM

	Training	Testing
Algorithm	accuracy	accuracy
Decision Tree Classifier	100.00%	100.00%
Random Forest Classifier	100.00%	100.00%
Gaussian NB	98.33%	99%
Logistic Regression	97.5%	98%
SVC Linear	95.8%	96%
Ada boost classifier	95%	95%
Gradient boosting classifier	95%	95%
KNN	94.14%	94.20%
SVC RBF	88.33%	88.33%

TABLE II
PRECISION, RECALL AND F1-SCORE OF EACH ALGORITHM

Algorithm	Precision	Recall	F1-Score
Decision Tree Classifier	1.00	0.92	0.96
Random Forest Classifier	1.00	0.99	0.99
Ada Boost Classifier	1.00	0.92	0.96
KNN	0.96	0.95	0.95
SVC Linear	0.99	0.95	0.97
Logistic Regression	1.00	0.92	0.96
SVC RBF	0.93	0.88	0.90
Gaussian NB	1.00	0.97	0.99

D. Model Evaluation and Selection

Based on the results (Table I, II), the algorithms which resulted in the highest accuracy in all 3 data sets were selected. Those are decision tree classifier, random forest classifier.

These models gave 100% of accuracy that helps majorly in predicting Chronic renal disease.

IV. DISCUSSION

This research gave 100% accurate result by random forest classifier and decision tree classifier. Here, some attributes like appetite, pedal edema, anemia, etc.. are not considered as they does not contribute towards prediction of CRD. Further, the missing values which were completely missed at random made it impossible to achieve a perfect accuracy without filling them from a collaborative imputer instead of a constant. Considering the medical importance of the attributes, some of them have a lesser co-relation compared to others because of the stage they appear in the patient. When training the models it makes a huge impact on the accuracy. After training the model, it clearly shows that tree structures have higher accuracy than other classification algorithms, which can be justified from the distribution of the data set since the selected attributes have a clearer separation in the class except for serum creatinine attribute. Therefore, from medical importance and the statistical methods and correlation the attributes are selected to give accurate prediction of CRD and finally got high accurate result with decision tree and random forest classifier.

V. CONCLUSION AND FUTURE WORK

Chronic renal disease (CRD) is a life-threatening issue that affects almost 14% of the world population and predicting it with a 100% overall accuracy makes it possible for people to get to know it in the early stages to get treated with a minimum cost and minimum risk. Proper feature engineering helps to reduce the number of features needed for the prediction algorithm and practically it reduces the number of medical tests to be taken. Filling missing values based on the distribution of them along with the collocation of other attributes by statistical methods like mean, median instead of replacing with a constant directly leads to higher accuracy in prediction models compared to the related work done with the same dataset. Furthermore, the decision trees classifier and the random forest classifier are the better algorithms to do the predictions for CRD since those have 100% overall accuracy This work suggests a new workflow including data preprocessing, missing values handling and features selection to predict CRD status as positive or negative. Furthermore, this work highlights the importance of incorporating the domain knowledge into feature selection when analysing clinical data related to CRD.

Accordingly, it is worthwhile to explore the use of KNN-imputer based approach to handle missing values in data sets related to multiple diseases in future. Furthermore, more insights into CRD can be gained by adding knowledge of family history, water consumption patterns and food types into the analysis.

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